Accuracy as a Fraction

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 26

Recall and choose your starting point

> Record your starting explanation for Accuracy as a Fraction before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Training and Testing

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Features and Distractors

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Debugging Strategies

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Accuracy as a Fraction trace and record the first step you can explain without help.

Use W2: Compare both class fractions for **Accuracy** and **class imbalance**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Accuracy as a Fraction explanation against the supplied facts before discussing it.

Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Assess a new comparison independently

New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Accuracy as a Fraction target to a trace and a justified conclusion.

Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two **percentage**s misleading here?

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Accuracy as a Fraction case and show the deciding evidence.

Nine of twelve held-out classifications are correct.

Compute **accuracy** and state the **denominator**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A classifier gets 9 of 12 cases right. Another gets 8 of 10 right. Calculate both accuracies and identify the larger fraction.

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 26

Project checkpoint

> Justify what Accuracy as a Fraction adds to Transparent Book Recommender and keep the evidence.

Report correct recommendations / all test users, alongside the raw test **count**.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two **percentage**s misleading here?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Score the majority baseline

Score the majority baseline. Apply the supplied Accuracy as a Fraction facts and keep your reasoning.

Training has 16 clear and 4 blocked labels. State the majority rule. On the worked evaluation, give its matrix, **accuracy**, false positives and false negatives with blocked positive. Explain why 80 percent is incomplete.

My reasoning and evidence

W2 Compare both class fractions

> Compare both class fractions. Apply the supplied Accuracy as a Fraction facts and keep your reasoning.

For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives. Compare with the baseline.

> **My reasoning and evidence**

W3 Explain a changed class mix

> Explain a changed class mix. Apply the supplied Accuracy as a Fraction facts and keep your reasoning.

For the illustrative 8-clear 8-blocked set, use baseline [[8,0],[8,0]] and candidate [[6,2],[0,8]]. Calculate ordinary and **balanced accuracy** for each. Explain what stayed fixed and what this example does not prove.

> **My reasoning and evidence**

WORKBOOK · 15 OF 26

W4 Assess a new comparison independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Assess a new comparison independently. Apply the supplied Accuracy as a Fraction facts and keep your reasoning.

New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

My reasoning and evidence

WORKBOOK · 16 OF 26

W5 Limit claims from missing classes

> Limit claims from missing classes. Apply the supplied Accuracy as a Fraction facts and keep your reasoning.

An evaluation contains ten clear cases and no blocked cases. All **prediction**s are clear. State ordinary **accuracy**, explain why blocked performance is unavailable, and describe how our code handles the missing class.

My reasoning and evidence

WORKBOOK · 17 OF 26

Summaries with counts and proportions — W1 State the denominator

Use the supplied Accuracy as a Fraction task to record your reasoning, evidence and next step.

In the worked poll, why is the known **denominator** 4 rather than 5?

My reasoning and evidence

WORKBOOK · 18 OF 26

Summaries with counts and proportions — W2 Compare proportions

> Use the supplied Accuracy as a Fraction task to record your reasoning, evidence and next step.

Poll A has 2 Yes of 4 known and Poll B has 3 Yes of 6 known. Compare **proportion**s and **count**s.

My reasoning and evidence

Summaries with counts and proportions — W3 Repair zero division

> Use the supplied Accuracy as a Fraction task to record your reasoning, evidence and next step.

All three responses are Unknown. What should the program report?

My reasoning and evidence

WORKBOOK · 20 OF 26

Summaries with counts and proportions — W4 Summarize independently

> Use the supplied Accuracy as a Fraction task to record your reasoning, evidence and next step.

Process Yes, Yes, No, Unknown, No, Yes. Report Yes **count**, known **count**, unknown **count** and Yes **proportion** among known.

My reasoning and evidence

WORKBOOK · 21 OF 26

Summaries with counts and proportions — W5 Avoid overclaiming

> Use the supplied Accuracy as a Fraction task to record your reasoning, evidence and next step.

Three of five known volunteers said Yes. Can you say 60 percent of all students agree?

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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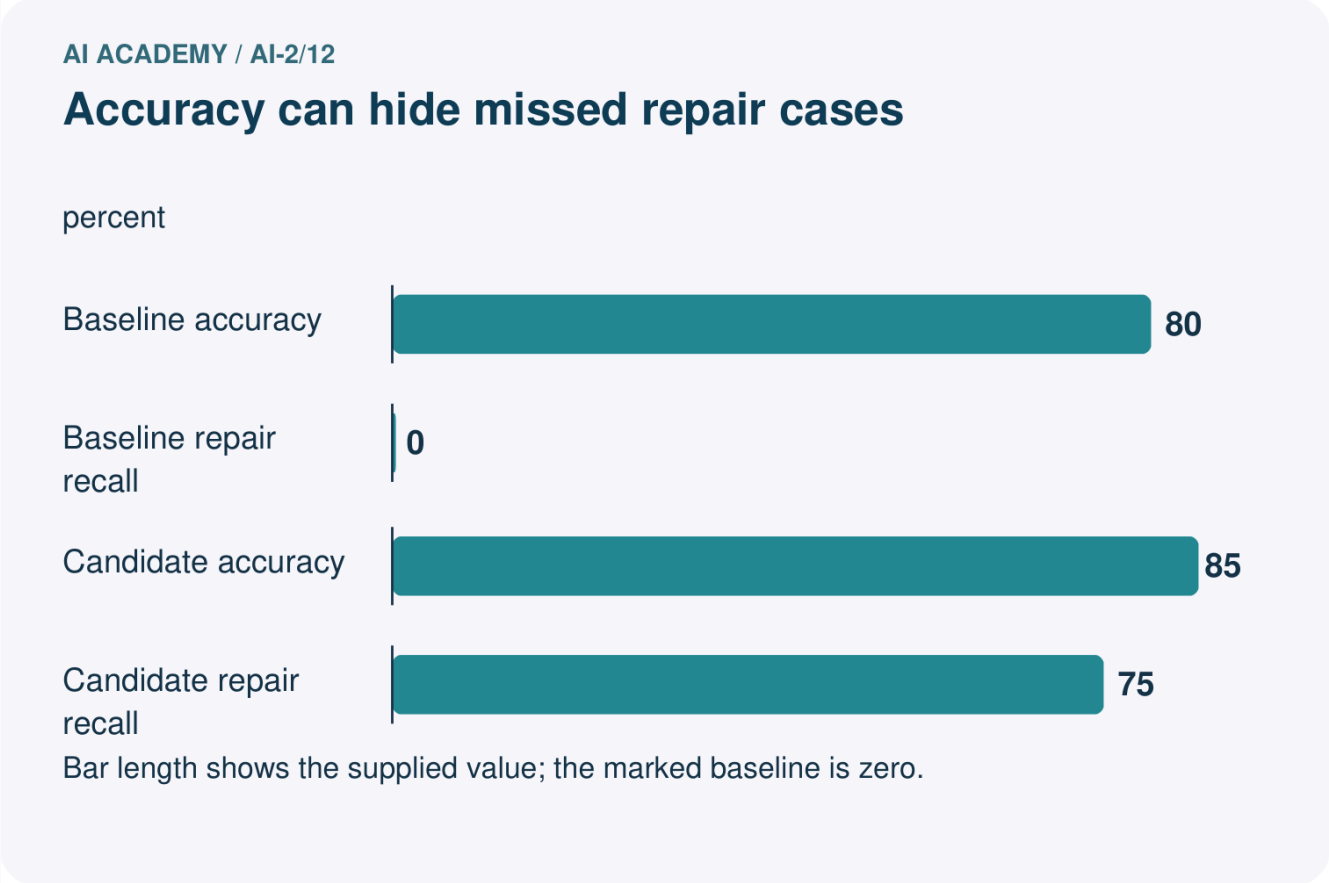
Pictorial case practice: Accuracy as a Fraction

> A fictional shelf classifier predicts repair needed or not needed. Explain the labelled step before checking the changed condition.

A fictional shelf classifier predicts repair needed or not needed.

Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed.

Always-not-needed baseline is correct on 16. Candidate has TP=3,FN=1,FP=2,TN=14.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/12

Accuracy as a Fraction

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Reference population

20 labelled cases, including four repair-needed books.

2. Baseline

16/20=80% accuracy; catches 0/4 needed cases.

3. Candidate

(3+14)/20=85% accuracy; catches 3/4=75% needed cases.

4. Compare costs

Candidate also raises two false alarms; discuss both missed repairs and extra checks.

CHANGE ONE THING

A candidate instead marks every book repair needed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is 100% repair detection not sufficient to declare the changed classifier best?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional shelf classifier predicts repair needed or not needed.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Candidate also raises two false alarms; discuss both missed repairs and extra checks.

1. Reference population

2. Compare costs

3. Candidate

4. Baseline

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare costs Candidate also raises two false alarms; discuss both missed repairs and extra checks.

A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A candidate instead marks every book repair needed. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Recall is 4/4=100%, but **accuracy** is only 4/20=20%.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Recall is 4/4=100%, but

> **accuracy** is only 4/20=20%. All 16 not-needed books become false alarms.

Repair a missing building block

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan **count**s?.

Return to this week and explain what you changed.

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the classifier always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

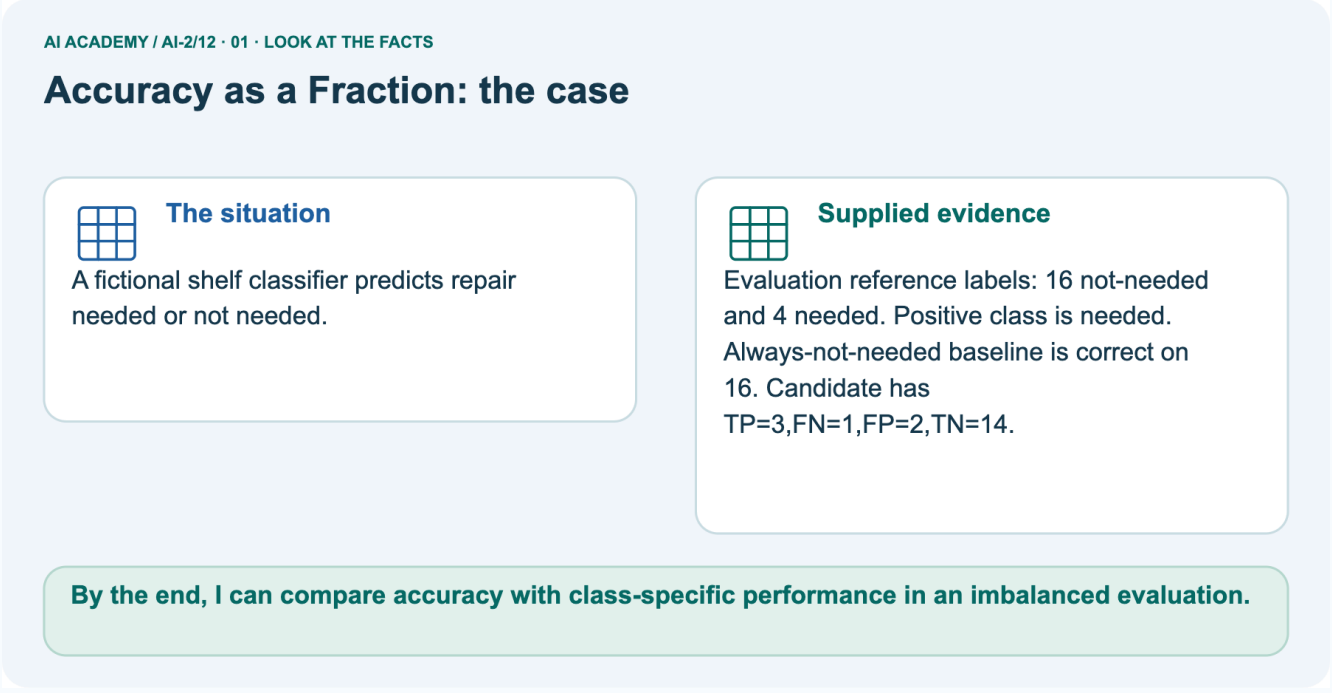
What I repaired, the evidence I kept and the next test

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each Accuracy as a Fraction picture from its labels, then record a relevant limit.

01. Accuracy as a Fraction: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

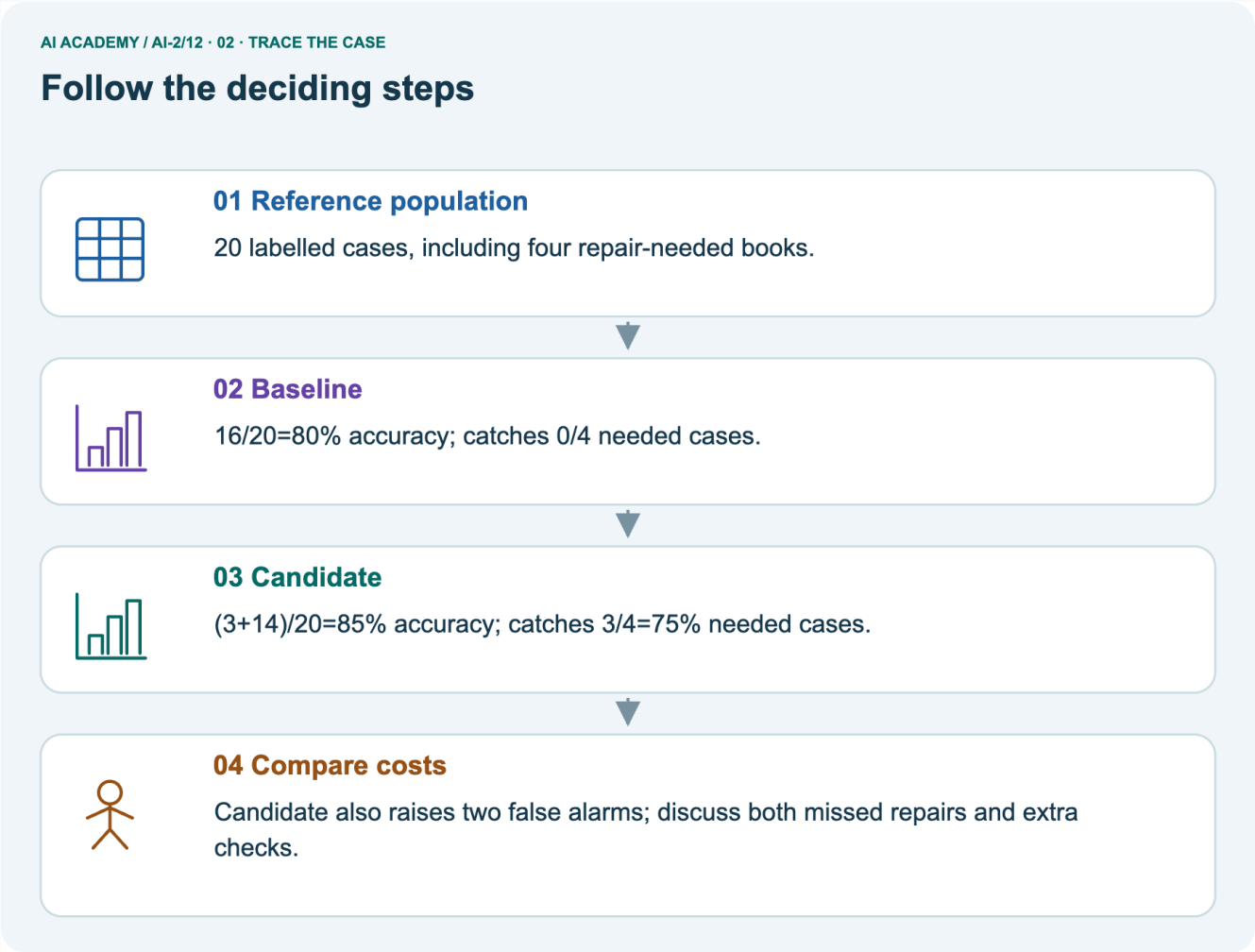
A fictional shelf classifier predicts repair needed or not needed. Supplied facts: Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed. Always-not-needed baseline is correct on 16.

Candidate has TP=3,FN=1,FP=2,TN=14.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



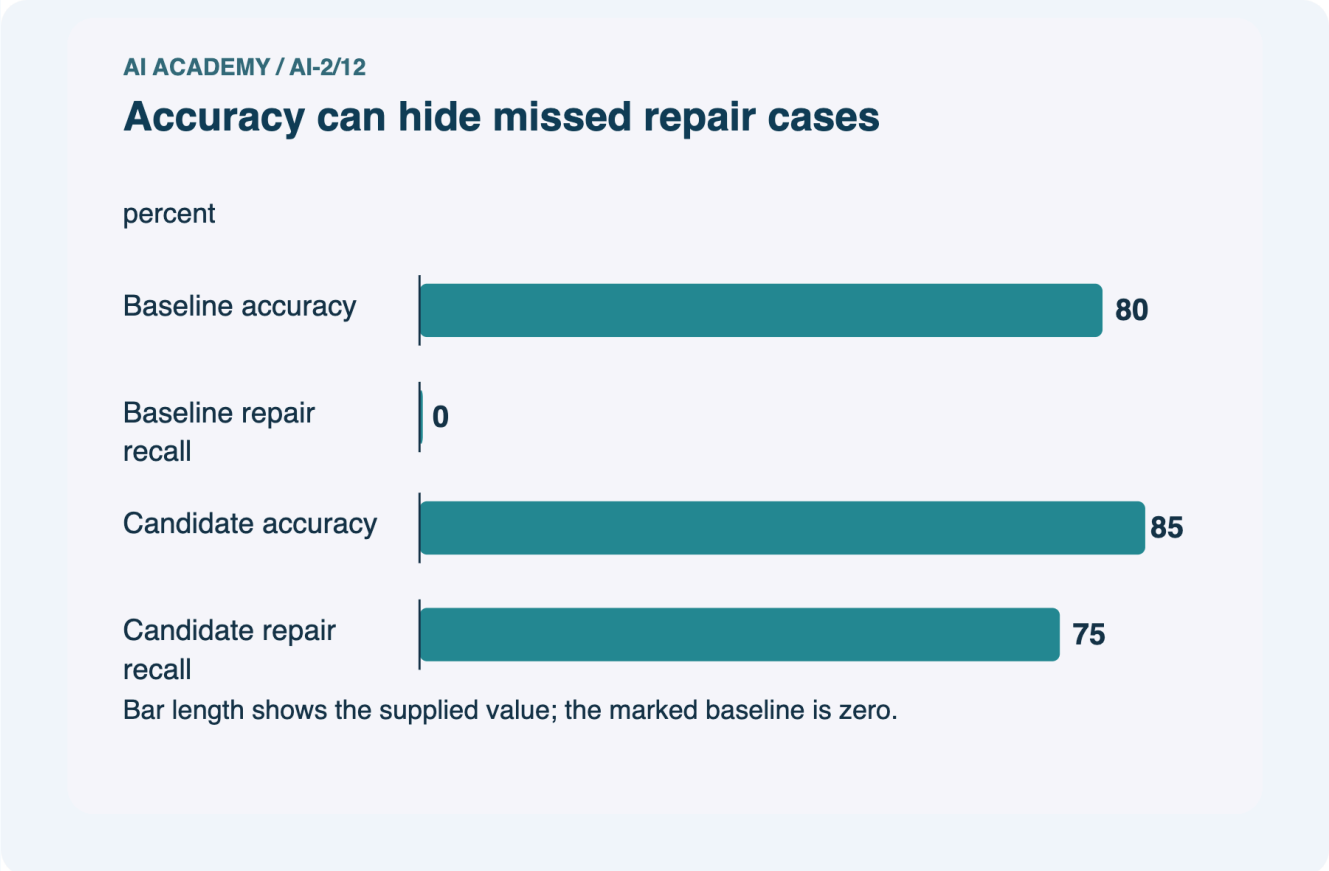
Follow the labelled picture with the matching explanation. Open the vector diagram

Reference population: 20 labelled cases, including four repair-needed books. → Baseline: 16/20=**80% accuracy**; catches 0/4 needed cases. → Candidate: (3+14)/20=85% **accuracy**; catches 3/4=75% needed cases. → Compare costs: Candidate also raises two false alarms; discuss both missed repairs and extra checks.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Accuracy can hide missed repair cases



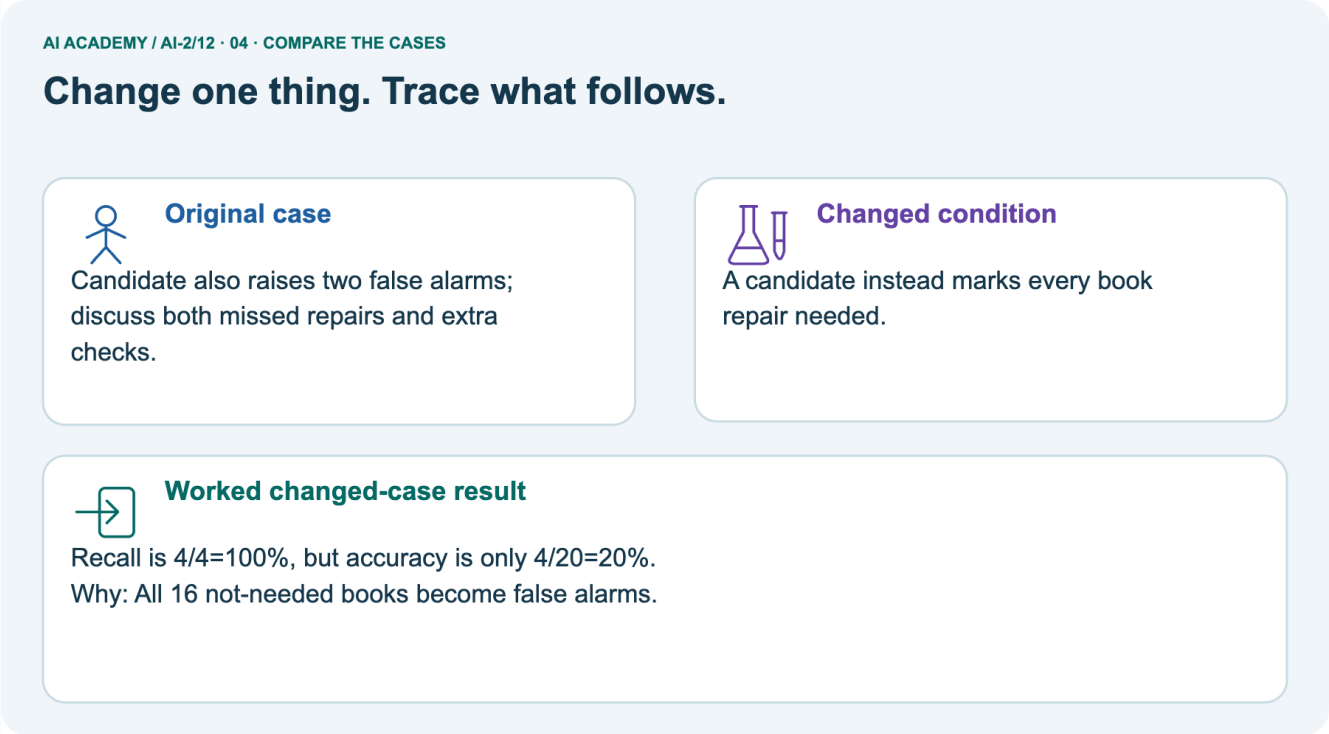
Follow the labelled picture with the matching explanation. Open the vector diagram

Accuracy can hide missed repair cases. Units: percent. Baseline **accuracy**: 80. Baseline repair recall: 0. Candidate **accuracy**: 85. Candidate repair recall: 75.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

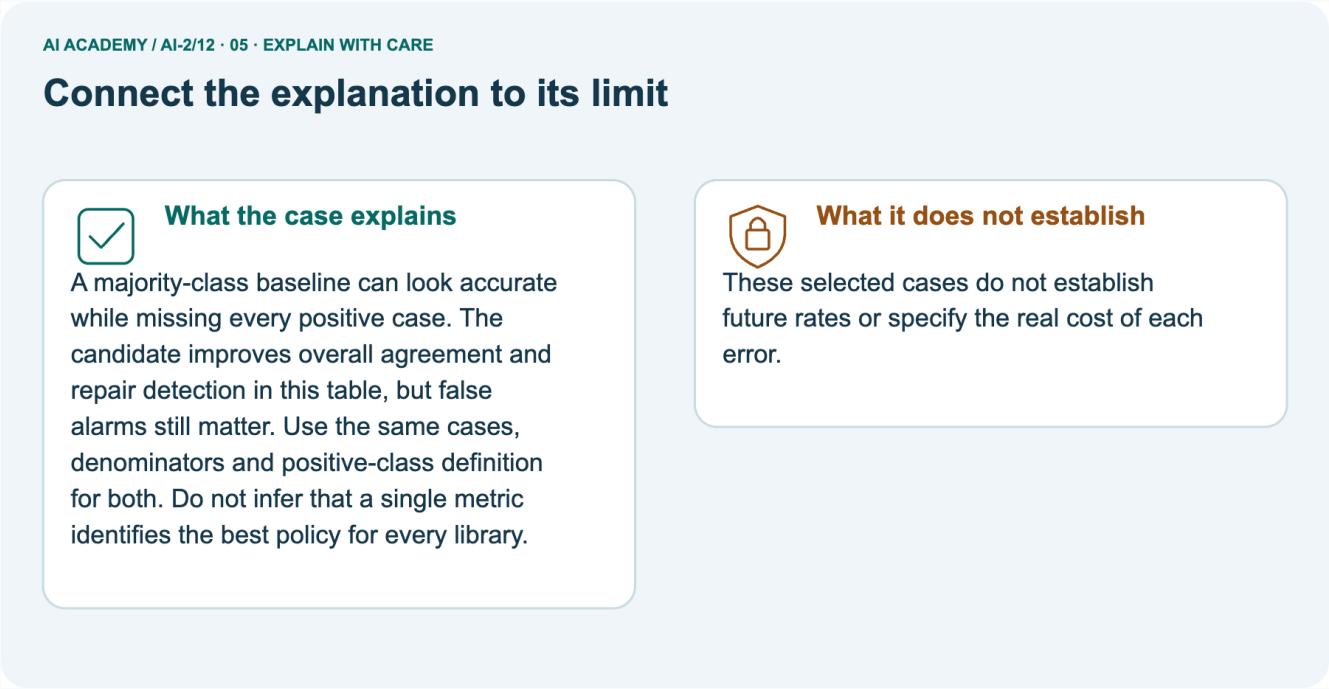
Original case: Candidate also raises two false alarms; discuss both missed repairs and extra checks. Changed condition: A candidate instead marks every book repair needed.

Expected result: Recall is 4/4=100%, but **accuracy** is only 4/20=20%. Reason: All 16 not-needed books become false alarms.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

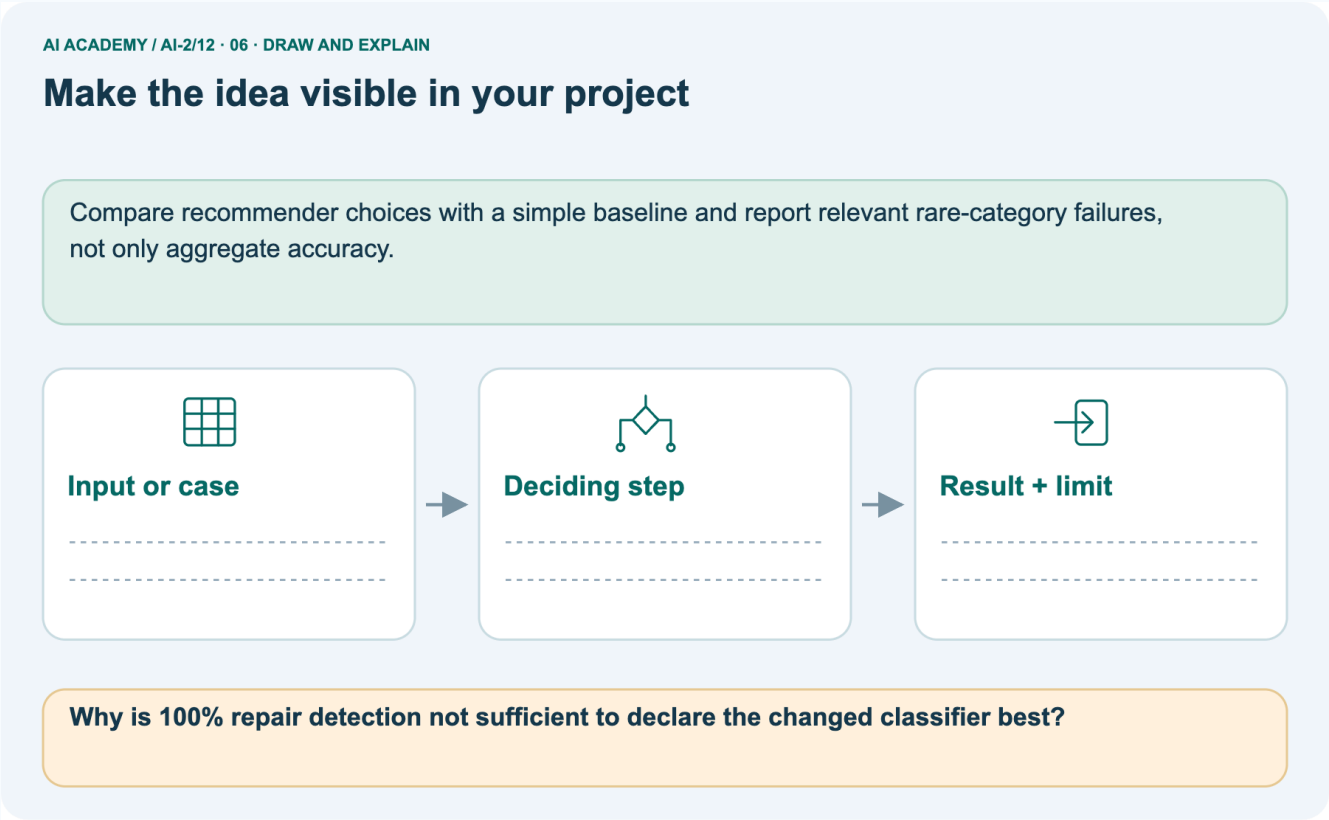
Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

Limit: These selected cases do not establish future rates or specify the real cost of each error.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**. Draw your **input**, deciding step, result and limit.

Supported question: Why is 100% repair detection not sufficient to declare the changed classifier best?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.